

## Experiment No. 5

### Experiment Name

Interactive Dashboard Development using Tableau / Microsoft Power BI

### Aim

To design and develop an interactive business dashboard using Tableau or Microsoft Power BI for analyzing real-world datasets and presenting key performance indicators (KPIs) to support data-driven decision-making.

### Purpose

Business Intelligence (BI) tools such as Tableau and Microsoft Power BI enable organizations to transform raw data into interactive dashboards and reports. Unlike static charts, dashboards allow users to filter, drill down, and explore data dynamically, helping managers monitor business performance and make informed decisions.

In this experiment, students will work with datasets such as the Superstore Sales Dataset, Global Superstore Dataset, Amazon Sales Dataset, or Netflix Dataset to create an interactive dashboard featuring KPIs, charts, filters, and slicers that provide meaningful business insights.

### Prerequisites

- Basic knowledge of data visualization concepts
- Familiarity with Tableau Public or Microsoft Power BI Desktop
- Understanding of data preprocessing and Exploratory Data Analysis (EDA)
- Cleaned dataset from previous experiments
- Basic knowledge of business metrics and KPIs

### Steps

1. Download and install Tableau Public or Microsoft Power BI Desktop.
2. Import a real-world dataset such as the Superstore Sales Dataset, Global Superstore Dataset, or Amazon Sales Dataset.
3. Verify the imported data by checking field names, data types, and relationships between tables (if applicable).
4. Create Key Performance Indicators (KPIs) such as Total Sales, Total Profit, Number of Orders, Average Order Value, or Customer Count.

5. Develop multiple visualizations, such as bar charts, line charts, pie charts, maps, and treemaps, to analyze business performance from different perspectives.
6. Add interactive filters, slicers, and drill-down features to allow users to explore the data by region, product category, customer segment, or time period.
7. Arrange all visualizations into a professional dashboard with appropriate titles, formatting, and layout.
8. Analyze the dashboard to identify business trends, top-performing products, high-revenue regions, and areas requiring improvement.
9. Publish or export the dashboard for presentation and sharing with stakeholders.
10. Prepare a brief report summarizing the key business insights obtained from the dashboard.

## Questions

1. What is Business Intelligence (BI)? How does it support organizational decision-making?
2. What is an interactive dashboard? How is it different from a static report?
3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.
4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.
5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.
6. What factors should be considered while designing an effective business dashboard?
7. How can dashboards help managers monitor organizational performance in real time?
8. Why is data visualization important in Business Intelligence tools?
9. What insights can be derived from a sales dashboard for an e-commerce company?
10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.